

Task 6: Password Strength Evaluation Report

Cybersecurity Internship – Elevate Labs

1. Objective

To understand what makes a password strong and evaluate multiple passwords using an online password strength tool (passwordmeter.com). This task demonstrates how password complexity directly affects security.

2. Tool Used

Tool: passwordmeter.com

Description: A real-time password strength analyzer that evaluates passwords based on NIST guidelines, time-to-crack estimates, and common pattern detection.

3. Passwords Tested & Results

Password	Strength	Time to Crack	Tool Feedback
password123	Very Weak	Instantly	Most commonly used password
Password1	Very Weak	Instantly	Most commonly used password
p@sswOrd	Very Weak	Instantly	Most commonly used password
Tr#9klmQ2	Fair	104M years (secure DB)	Uses all 4 character types; needs 12+ chars
X#7vb!qN2\$mPw@3	Strong	> 1 Trillion years	Good length, all char types, no patterns

4. Observations

From the test results, the following patterns were observed:

- Simple passwords like 'password123' and 'Password1' are cracked instantly — they appear in common password databases.
- Substituting letters with symbols (e.g., p@sswOrd) does NOT significantly improve strength if the base word is common.
- Using all four character types (uppercase, lowercase, numbers, symbols) significantly improves strength.
- Length is the most critical factor — passwords with 12+ characters are exponentially harder to crack.
- Random, pattern-free passwords like 'X#7vb!qN2\$mPw@3' achieve maximum security ratings.

5. Best Practices for Strong Passwords

- Use a minimum of 12 characters (16+ recommended).
- Mix uppercase letters, lowercase letters, numbers, and special symbols.
- Avoid dictionary words, names, or personal information (birthdays, names, etc.).
- Never reuse passwords across multiple accounts.
- Use a password manager (e.g., Bitwarden, LastPass) to generate and store strong passwords.
- Consider using passphrases: e.g., 'Mango\$Runs@Fast99' — long yet memorable.

6. Common Password Attacks

Brute Force Attack

An attacker systematically tries every possible combination of characters. Short passwords (under 8 chars) can be cracked in seconds with modern hardware.

Dictionary Attack

Uses a predefined list of common words, names, and passwords. Passwords like 'password', 'admin', or 'Password1' are instantly matched.

Credential Stuffing

Attackers use leaked username/password pairs from data breaches to attempt login on other services — highlighting why unique passwords matter.

7. Conclusion

Password complexity has a direct and measurable impact on security. Even small improvements — such as adding symbols or increasing length — can change crack time from 'instantly' to 'millions of years'. The strongest password tested (X#7vb!qN2\$mPw@3) would take over 1 trillion years to crack online, demonstrating that randomness and length are the most powerful defenses against password attacks.